

Feedback on headless-macs

Hi Michael,

I run a pair of headless Apple Silicon Macs as inference nodes in my homelab and went through your repo in detail while comparing it against my own setup. It's well-built: the idempotent apply pattern, precheck/verify/restore lifecycle, and especially `known-issues.md` and `ram-sizing.md` are better operational reference than most of what's published on this topic. The FileVault headless-reboot catch alone is worth the read, and I've adopted several of your settings into my own config (disablesleep, standby/autopoweroff, womp, the caffeinate safety net, Spotlight off).

A few things I found that you may want to fix or consider, roughly in priority order:

1. `sysctl` persistence is likely broken

`setup.sh` persists network tuning by appending to `/etc/sysctl.conf`, but macOS stopped reading that file at boot somewhere around Catalina. The `sysctl -w` calls apply for the running session, then everything silently reverts on reboot, so `verify.sh` passes right after setup but the tuning is gone after the first restart.

Fix: a LaunchDaemon with `RunAtLoad` that runs `/usr/sbin/sysctl -w` for each key. You already have the plist-install pattern in the repo, so it's a small change.

References:

- <https://discussions.apple.com/thread/253840320>
- <https://gist.github.com/pythoninthegrass/8073e5e3b24f385c9d9b712f6f243982>

2. `pmset autorestart 1` is missing from `setup.sh`

Restart-after-power-failure is essential for a 24/7 headless node; without it, any power blip means a truck roll. It appears only in the deprecated `pmset_to_ollama.sh`, and there it's set to 0. Suggest `pmset_apply autorestart 1` in the power section of `setup.sh`.

3. Default network posture is wide open

`config.json` defaults to `localhost_only: false` (everything binds 0.0.0.0) combined with `disable_firewall: true`. On a typical home LAN that exposes unauthenticated Ollama/mlx-lm/Infinity APIs to every device on the network, and Ollama's API can pull/delete models, not just infer.

Suggestion: instead of disabling the firewall, ship a pf anchor that allows inbound on the serving ports only from an allowlist of client IPs (plus localhost), default-deny otherwise. I run exactly this on my nodes and it works fine with launchd-managed daemons. At minimum, flip the default to `disable_firewall: false` and document the allowlist as the recommended path.

4. Daemons run as root to work around the `$HOME` crash

Your known-issues doc handles the `panic: $HOME is not defined` crash by setting `HOME=/var/root` and running the serving daemons as root. The alternative that avoids root-owned network-facing daemons entirely: set `UserName` to an unprivileged user in the LaunchDaemon plist and set `HOME` to that user's home in `EnvironmentVariables`. The daemon still starts at boot with no GUI login, `OLLAMA_MODELS` is honored, and a compromise of the serving process doesn't hand over root.

5. sshd hardening via sed on the main config

`setup.sh` edits `/etc/ssh/sshd_config` in place. macOS updates can replace that file, silently dropping your hardening. Modern macOS ships an `Include /etc/ssh/sshd_config.d/*` directive, so a drop-in file (e.g. `/etc/ssh/sshd_config.d/100-headless.conf`) survives OS updates and is easier to make idempotent than the sed/append logic. Bonus: macOS sshd is launchd socket-activated, so new connections pick up the drop-in without the stop/start dance.

One more note on the hardening itself: setting `PasswordAuthentication no` before a key login is confirmed working is a lockout risk on a headless box. Might be worth a precheck gate (refuse to harden unless an `authorized_keys` file exists) or an explicit config flag.

6. Consider raising maxfiles

Concurrent model loading plus parallel inference connections can exhaust the default file-descriptor limit. A `limit.maxfiles` LaunchDaemon (`launchctl limit maxfiles 524288 1048576`) is cheap insurance and fits your existing plist pattern.

7. The pmset-reset-after-update problem could self-heal

known-issues correctly documents that macOS updates reset pmset values, with "re-run setup.sh" as the fix. Since the scripts are already idempotent, a launchd timer (daily `StartCalendarInterval`) re-running the system-baseline section would close that gap automatically. That's effectively what I do, just with ansible-pull on a 15-minute timer; your bash would work just as well on a daily one.

Happy to turn any of these into PRs if useful. Thanks for publishing this — the docs saved me real time.

Jeff